

PATARLAGELE, Romania

WMO: 153280

Lat: 45.3247N

Lon: 26.3694E

Elev: 953

StdP: 14.20

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.6	16.8	6.6	8.0	18.8	10.2	9.6	22.0	28.1	34.6	23.4	33.5	2.8	350	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	88.8	70.4	86.3	69.7	83.8	68.9	73.9	84.7	72.0	82.6	70.4	81.0	4.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.7	117.0	79.7	68.4	108.1	77.6	66.6	101.5	76.2	38.1	83.9	36.3	82.8	34.9	81.3	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	14.4	12.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.6	31.0	36.1	42.1	50.3	60.5	68.0	70.0	70.4	61.8	51.0	42.1	35.3		
	DBStd	15.37	7.75	7.55	8.29	7.42	5.70	5.03	5.15	5.40	7.05	7.25	7.67	7.45		
	HDD50	2132	588	391	267	86	4	0	0	0	3	75	259	459		
	HDD65	5420	1053	810	711	442	160	26	17	17	142	433	687	922		
	CDD50	2728	0	1	22	95	329	541	619	633	358	107	22	1		
	CDD65	541	0	0	0	1	20	117	170	185	47	1	0	0		
	CDH74	5063	0	0	2	57	258	1072	1502	1653	475	43	1	0		
	CDH80	1571	0	0	0	5	37	287	499	608	133	2	0	0		
(14) Wind	WSAvg	4.1	4.6	4.5	4.7	4.5	3.9	3.9	3.6	3.9	3.7	3.4	3.7	4.3		
(15) Precipitation	PrecAvg	26.4	1.2	1.0	1.3	1.9	3.1	4.0	3.7	3.0	2.2	2.0	1.5	1.5		
	PrecMax	36.9	3.4	3.0	2.6	4.6	6.7	6.5	9.2	6.7	4.5	6.2	3.8	4.0		
	PrecMin	17.3	0.2	0.1	0.1	0.8	1.0	1.5	1.2	0.7	0.4	0.1	0.1	0.2		
	PrecStd	5.2	0.7	0.7	0.6	1.0	1.5	1.2	1.9	1.6	1.1	1.3	0.9	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.0	64.7	72.0	79.8	84.1	89.0	92.2	91.7	88.3	79.1	69.4	61.0		
		MCWB	46.6	52.3	55.7	59.7	66.1	72.9	74.1	70.3	67.4	61.1	56.2	49.6		
	2%	DB	51.0	58.2	66.2	74.6	79.4	86.3	88.2	89.0	83.3	74.2	62.8	54.0		
		MCWB	43.4	47.6	52.1	58.6	64.1	70.6	71.9	69.1	65.3	59.0	52.5	45.8		
	5%	DB	46.9	53.1	61.7	69.8	76.7	83.5	85.4	86.4	78.9	69.5	57.5	50.1		
		MCWB	41.1	45.2	49.8	55.6	62.8	69.7	71.1	68.6	63.9	57.2	50.5	44.8		
	10%	DB	42.4	48.3	56.5	65.2	73.4	80.5	82.6	83.3	74.7	64.2	53.4	46.3		
		MCWB	38.2	42.3	47.5	53.5	61.1	68.2	69.8	67.7	62.6	54.1	48.3	41.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.9	54.0	58.4	61.8	71.5	75.6	76.9	74.5	70.2	62.7	57.1	50.8		
		MCDB	55.4	62.9	69.7	75.5	81.6	86.3	86.0	88.9	83.4	75.1	68.0	57.7		
	2%	WB	44.5	49.0	53.6	59.7	66.6	72.8	74.5	72.2	67.4	60.1	54.4	47.7		
		MCDB	50.1	57.7	65.1	73.3	74.9	83.8	84.3	84.8	80.0	72.4	61.3	53.4		
	5%	WB	41.3	45.6	50.6	57.5	64.4	70.7	72.5	70.7	65.4	57.8	51.7	44.7		
		MCDB	45.4	51.9	59.9	68.7	73.7	81.7	81.9	82.4	76.7	68.4	56.4	50.2		
	10%	WB	39.0	42.8	47.9	55.1	62.4	69.0	70.8	69.1	63.5	55.0	49.2	42.1		
		MCDB	42.2	47.6	55.9	64.3	71.6	79.4	80.2	80.3	73.7	62.9	53.0	46.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.0	18.1	19.4	20.6	21.0	20.7	21.1	22.7	21.0	20.8	16.0	14.0		
		MCDBR	21.8	26.8	29.5	30.4	26.4	24.8	24.7	26.7	26.8	29.9	23.9	21.2		
		MCWBR	15.4	17.1	16.4	15.4	13.5	12.2	11.7	11.4	12.2	16.2	14.4	14.2		
	5% WB	MCDBR	21.1	26.2	28.6	28.7	24.5	23.3	21.8	24.0	25.1	29.3	22.2	21.1		
		MCWBR	14.7	16.9	16.1	14.4	13.9	12.5	11.7	11.3	12.0	16.1	13.8	13.6		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.350	0.389	0.450	0.426	0.429	0.443	0.458	0.415	0.377	0.348	0.321		
	taud		2.340	2.253	2.212	2.069	2.215	2.242	2.211	2.161	2.247	2.337	2.368	2.359		
	Ebn at Noon		246	259	264	257	268	267	261	252	253	246	233	232		
	Edh at Noon		25	33	39	49	44	43	44	44	37	30	24	22		
(44) All-Sky Solar Radiation	RadAvg		418	679	1044	1421	1780	1897	1931	1742	1267	830	458	350		
	RadStd		52	84	110	148	145	144	143	118	111	96	47	45		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)	+1.01	N/A	N/A	N/A	+1.19	N/A	N/A	-212	-290	+137	+52

Nomenclature: See separate page